

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 2- Case Study (2 QUESTIONS)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 – Manipulation (BL2)	Assessment:	Assessment Tool 1 – Case Study
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10Marks)
CO1 Statement:	Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)		
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Section / Batch:	BTECH AIADS	Date:	21.07.2026

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Code of Conduct

I RENEE VINCENT ASR-1924 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renée Vincent

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Aesthetic Mapping & Dashboard Design	Mapping categorical, ordinal, and quantitative data to position, size, shape, and color; selection of scales, axes, and coordinate systems for business dashboards	1	10
Nonlinear Axes & Health Data Visualization	Use of logarithmic scales for exponential data, time-series representation, coordinate systems, and color scales for highlighting high-risk regions	2	10
GRAND TOTAL:			20 Marks

Annexure A –Case Study

1.A multinational retail company operates hundreds of stores across different regions and collects data on product categories, monthly sales, customer satisfaction levels, profit margins, and promotional campaign effectiveness. The management plans to develop an interactive analytics dashboard that enables regional managers to compare store performance, identify top-selling products, monitor sales trends, evaluate customer satisfaction, and quickly detect areas requiring attention. As a data visualization specialist, explain how you would design this dashboard by mapping categorical, ordinal, and quantitative data onto appropriate visual aesthetics. Discuss the selection of scales, coordinate systems, and axes for accurate representation and comparison of data. Furthermore, explain how color can be used to distinguish product categories, represent performance metrics, and highlight critical business issues. Justify your choice of color palettes and visualization techniques to ensure clarity, accessibility, and effective decision-making.

2.A national health agency is monitoring the spread of a contagious disease across multiple states using data collected from hospitals and healthcare centers. The dataset includes daily infection counts, mortality rates, vaccination coverage, population density, and disease growth rates, with some regions experiencing exponential increases in infections. Public health officials require an effective visualization system to identify outbreak hotspots, track disease progression over time, allocate medical resources efficiently, and communicate risks to the public. As a data visualization consultant, propose a visualization strategy that appropriately maps different data types onto visual aesthetics and utilizes suitable scales, coordinate systems, and axes for meaningful interpretation. Discuss the use of nonlinear axes for handling highly skewed infection data, and explain how color scales can represent disease severity while highlighting high-risk regions. Justify the selection of accessible and effective color palettes and evaluate how your visualization approach supports timely and informed public health decision-making.

1) Designing an Interactive Retail Analysis

Dashboard:

An interactive analytics dashboard should present business data in a clear, accurate and visually appealing manner to help regional managers make informed decisions.

A) Mapping Data to Visual Aesthetics:

Data Type	Example	Visual Aesthetic	Recommended chart
Categorical	Product Category, Region store.	Colours, shape	Bar chart, Pie chart
Ordinal	Customer Satisfaction (Low, Medium, High), Store Ranking	ordered position, size	ordered Bar chart
Quantitative	Monthly Sales, Profit, Margin, Campaign Revenue	Position, Length, Height	Line chart, Column chart, Scatter Plot.

B) Selection of Scales:

- * Linear Scale: Used for monthly sales, profit margins and revenue.
- * Ordinal Scale: Used for customer satisfaction levels and store rankings.
- * Categorical Scale: Used for product categories and regions.
- * Percentage Scale: Used for profit margins and campaign effectiveness.

C) Coordinate Systems And Axes:

- * Cartesian Coordinate System: Used for line charts, bar & scatter plots.
- * X-axis: Time, product categories or regions.
- * Y-axis: Sales, Profit margin, customer satisfaction score or revenue.
- * Consistent Axis Scale: Enables fair comparison across stores and regions.
- * Interactive Filters: Allow users to select region, store category and time period.

D) Use of colour:

- * Electronics - Blue
- * Clothing - Green
- * Grocery - Orange
- * Furniture - Purple
- * Beauty - Pink

} → Product categories

Performance Metrics :

* Light Green → Dark Green

Critical Business Issues :

* Red → Poor

* Yellow → Moderate

* Green → Excellent

E) Visualization Techniques :

* Line Chart

* Grouped Bar chart

* Stacked Bar chart

* Scatter Plot

* Heat Map

* KPI Cards

* Interactive Filters

* Tooltips & Drill-down

F) Justification of Colour Palettes :

* Qualitative: Best for distinguishing.

* Sequential: Increasing or decreasing values.

* Diverging: Highlights performance relative to targets by emphasizing positive & negative values.

* Colour Blind Friendly Colours: Improve accessibility and reliability for all users.

G) Benefits :

* Enables quick comparison of store performance.

* Identifies sales trends and top selling products.

* Highlights critical business issues instantly.

* Supports faster data driven decision making.

Conclusion :

An interactive dashboard with effective visualizations and colour coding enables managers to quickly analyze retail performance & make informed business decisions.

2) Visualization Strategy for Disease Monitoring Dashboard:

A) Mapping Data to Visual Aesthetics:

Data Type	Example	Visual Aesthetic	Visualization
Categorical	State, Region	colour, shape	Map, Bar chart
Ordinal	Disease Severity, (low, medium, high)	ordered colour intensity	Heat Map
Quantitative	Infection count, Mortality Rate, Vaccination Coverage, Population Density	Position, Length, size	line chart, Scatter Plot

B) Scales, coordination systems and Axes:

- * Linear Scale: Vaccination coverage and mortality rate.
- * Logarithmic Scale: Daily infection counts and disease growth rates to display exponential increases clearly.
- * Cartesian Coordinate System: Used for line, bar & scatter plots.
- * Geographic coordinate System: Used for choropleth maps to display state-wise outbreaks.
- * x-axis: Time.
- * y-axis: Infection counts, mortality rate or vaccination percentage.

Nonlinear Axes:

Use a logarithmic Scale for infection counts to effectively visualize regions with exponential growth and compare both low & high infection values without distortion.

D) Use of colour:

- * Sequential: Represents increasing disease severity.
- * Diverging: Shows low, medium and high risk regions.
- * Blue: Represents vaccination coverage, with darker blue indicating higher coverage.

E) Visualization Techniques:

- | | |
|------------------|-----------------------|
| * Choropleth Map | * Heat Map |
| * Line chart | * Scatter plot |
| * Bar chart | * Interactive filters |

F) Justification of Colour Palettes:

- * Sequential colours clearly indicating increasing severity.
- * Diverging colours emphasize safe and high risk regions.
- * Colour blind friendly palettes improve accessibility and reliability.

G) Benefits:

- * Quickly identifies outbreak hotspots.
- * Track disease spread and vaccination progress.
- * Supports efficient medical resource allocation.
- * Enables faster public health decisions.

H) Conclusion:

A well designed visualization dashboard with appropriate scales, maps & colour coding enables rapid disease monitoring, effective resource planning and timely public health decision making.

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